

ECON3200 Tutorial 8 Handout

Money in the Short Run: Shocks to the New Keynesian Economy

Teresa Chang

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1. Lecture Recap for Tutorial 8

Tutorial 8 applies the three-equation New Keynesian model to short-run shocks. The three core equations are:

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t,$$

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t,$$

$$\hat{i}_t = \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t.$$

The first equation is the **New Keynesian IS curve**. It represents the demand side. A higher nominal interest rate raises the real interest rate and reduces current demand:

$$\hat{i}_t \uparrow \Rightarrow \hat{x}_t \downarrow.$$

The second equation is the **New Keynesian Phillips curve**. It represents firm price-setting. A positive output gap creates upward pressure on inflation:

$$\hat{x}_t \uparrow \Rightarrow \hat{\pi}_t \uparrow.$$

The third equation is the **Taylor rule**. It represents monetary policy. The central bank raises the nominal interest rate when inflation or the output gap increases:

$$\hat{\pi}_t \uparrow \Rightarrow \hat{i}_t \uparrow, \quad \hat{x}_t \uparrow \Rightarrow \hat{i}_t \uparrow.$$

1.1 Output Gap, Output, and Labour

The output gap is defined as

$$\hat{x}_t = \hat{y}_t - \hat{y}_t^f.$$

Therefore,

$$\hat{y}_t = \hat{x}_t + \hat{y}_t^f.$$

The production function in the New Keynesian model is

$$\hat{y}_t = \hat{z}_t + \hat{l}_t.$$

Therefore, labour is

$$\hat{l}_t = \hat{y}_t - \hat{z}_t.$$

This formula says that output can increase either because productivity rises or because labour rises. So after solving for output and productivity, labour is the remaining part:

$$\text{labour} = \text{output} - \text{productivity}.$$

1.2 Key Parameters Used in Tutorial 8

Throughout Tutorial 8, the parameter values are:

$$\beta = 0.75, \quad \sigma = \phi = 1, \quad \delta_\pi = 2, \quad \delta_x = 0.5, \quad \kappa = 0.5.$$

The slope of the New Keynesian Phillips curve is

$$\tilde{\kappa} = (\sigma + \phi) \frac{(1 - \kappa)(1 - \beta\kappa)}{\kappa}.$$

Substitute the values:

$$\tilde{\kappa} = (1 + 1) \frac{(1 - 0.5)(1 - 0.75 \times 0.5)}{0.5}.$$

$$\tilde{\kappa} = 2 \frac{0.5(0.625)}{0.5} = 1.25.$$

Therefore, in this tutorial,

$$\hat{\pi}_t = 0.75E_t[\hat{\pi}_{t+1}] + 1.25\hat{x}_t.$$

Memorable Tip

The main solving strategy is:

$$\text{Shock} \Rightarrow \hat{y}_t^f \text{ or shock term} \Rightarrow \hat{x}_t \Rightarrow \hat{\pi}_t \Rightarrow \hat{i}_t \Rightarrow \hat{y}_t \Rightarrow \hat{l}_t.$$

2. Exercise 8.1: Negative Productivity Shock

Tutorial Question

Consider the New Keynesian model described by

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t, \quad (1.NK^*)$$

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t, \quad (2.NK^*)$$

$$\hat{i}_t = \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t. \quad (3.NK^*)$$

with

$$\tilde{\kappa} = (\sigma + \phi) \frac{(1 - \kappa)(1 - \beta\kappa)}{\kappa},$$

$$v_t = \rho_v v_{t-1} + \varepsilon_t, \quad u_t = E_t[\hat{y}_{t+1}^f] - \hat{y}_t^f,$$

$$\hat{y}_t^f = \left(\frac{1 + \phi}{\sigma + \phi} \right) \hat{z}_t, \quad \hat{z}_t = a_t, \quad a_t = \rho_a a_{t-1} + \omega_t.$$

Let the economy be at steady state in period 0. Assume that there is an unanticipated negative productivity shock in period 1, i.e.

$$\omega_1 = -1,$$

and that there are no other shocks in the economy. Let

$$\beta = 0.75, \quad \sigma = \phi = 1, \quad \delta_\pi = 2, \quad \delta_x = 0.5, \quad \rho_a = 0, \quad \kappa = 0.5.$$

All expectations coincide with true values except at period 1. The shock dies off at period 2, so

$$E_1[\hat{x}_2] = \hat{x}_2 = 0, \quad E_1[\hat{\pi}_2] = \hat{\pi}_2 = 0.$$

Compute

$$\hat{i}_1, \quad \hat{x}_1, \quad \hat{\pi}_1, \quad \hat{y}_1, \quad \hat{l}_1.$$

Solution

For these parameter values, the three New Keynesian equations become

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \hat{i}_t + E_t[\hat{\pi}_{t+1}] + u_t,$$

$$\hat{\pi}_t = 0.75E_t[\hat{\pi}_{t+1}] + 1.25\hat{x}_t,$$

$$\hat{i}_t = 2\hat{\pi}_t + 0.5\hat{x}_t.$$

There are no monetary policy shocks, so

$$v_t = 0.$$

The productivity process is

$$a_t = \rho_a a_{t-1} + \omega_t.$$

Since the economy starts at steady state,

$$a_0 = 0.$$

Given

$$\rho_a = 0, \quad \omega_1 = -1,$$

we get

$$a_1 = 0(0) + (-1) = -1.$$

Since

$$\hat{z}_t = a_t,$$

we have

$$\hat{z}_1 = -1.$$

Flexible-price output is

$$\hat{y}_t^f = \left(\frac{1 + \phi}{\sigma + \phi} \right) \hat{z}_t.$$

Since $\sigma = \phi = 1$,

$$\frac{1 + \phi}{\sigma + \phi} = \frac{2}{2} = 1.$$

Therefore,

$$\hat{y}_1^f = \hat{z}_1 = -1.$$

Since the shock dies off at period 2,

$$E_1[\hat{y}_2^f] = 0.$$

Now compute the IS shock term:

$$u_1 = E_1[\hat{y}_2^f] - \hat{y}_1^f.$$

Substitute:

$$u_1 = 0 - (-1) = 1.$$

So

$$u_1 = 1.$$

In period 2, all variables are at steady state:

$$\hat{x}_2 = \hat{\pi}_2 = \hat{i}_2 = 0.$$

Therefore, in period 1,

$$E_1[\hat{x}_2] = 0, \quad E_1[\hat{\pi}_2] = 0.$$

Use the IS curve:

$$\hat{x}_1 = E_1[\hat{x}_2] - \hat{i}_1 + E_1[\hat{\pi}_2] + u_1.$$

Substitute:

$$\hat{x}_1 = 0 - \hat{i}_1 + 0 + 1.$$

Therefore,

$$\hat{x}_1 = -\hat{i}_1 + 1.$$

Use the Phillips curve:

$$\hat{\pi}_1 = 0.75E_1[\hat{\pi}_2] + 1.25\hat{x}_1.$$

Since $E_1[\hat{\pi}_2] = 0$,

$$\hat{\pi}_1 = 1.25\hat{x}_1.$$

Substitute $\hat{x}_1 = -\hat{i}_1 + 1$:

$$\hat{\pi}_1 = 1.25(-\hat{i}_1 + 1).$$

Use the Taylor rule:

$$\hat{i}_1 = 2\hat{\pi}_1 + 0.5\hat{x}_1.$$

Substitute $\hat{\pi}_1 = 1.25(-\hat{i}_1 + 1)$ and $\hat{x}_1 = -\hat{i}_1 + 1$:

$$\begin{aligned}\hat{i}_1 &= 2 \left[1.25(-\hat{i}_1 + 1) \right] + 0.5(-\hat{i}_1 + 1) \\ &= 2.5(-\hat{i}_1 + 1) + 0.5(-\hat{i}_1 + 1) \\ &= 3(-\hat{i}_1 + 1) \\ &= -3\hat{i}_1 + 3.\end{aligned}$$

Therefore,

$$4\hat{i}_1 = 3.$$

So

$$\hat{i}_1 = 0.75.$$

Now recover the output gap:

$$\hat{x}_1 = -\hat{i}_1 + 1 = -0.75 + 1 = 0.25.$$

Inflation is

$$\hat{\pi}_1 = 1.25\hat{x}_1 = 1.25(0.25) = 0.3125.$$

Actual output is

$$\hat{y}_1 = \hat{x}_1 + \hat{y}_1^f.$$

Substitute:

$$\hat{y}_1 = 0.25 + (-1) = -0.75.$$

Labour is

$$\hat{l}_1 = \hat{y}_1 - \hat{z}_1.$$

Substitute:

$$\hat{l}_1 = -0.75 - (-1) = 0.25.$$

Therefore,

$$\boxed{\hat{i}_1 = 0.75, \quad \hat{x}_1 = 0.25, \quad \hat{\pi}_1 = 0.3125, \quad \hat{y}_1 = -0.75, \quad \hat{l}_1 = 0.25.}$$

Economic Interpretation

The negative productivity shock lowers productivity:

$$\hat{z}_1 = -1.$$

This lowers flexible-price output:

$$\hat{y}_1^f = -1.$$

However, actual output does not fall as much as flexible-price output:

$$\hat{y}_1 = -0.75, \quad \hat{y}_1^f = -1.$$

Since

$$-0.75 > -1,$$

actual output is above flexible-price output. Therefore, the output gap is positive:

$$\hat{x}_1 = \hat{y}_1 - \hat{y}_1^f = -0.75 - (-1) = 0.25 > 0.$$

A positive output gap raises inflation:

$$\hat{\pi}_1 = 0.3125 > 0.$$

Since inflation and the output gap are both positive, the Taylor rule implies that the nominal interest rate rises:

$$\hat{i}_1 = 0.75 > 0.$$

$$\text{Productivity} \downarrow \Rightarrow \hat{y}_1^f \downarrow \Rightarrow \hat{y}_1 > \hat{y}_1^f \Rightarrow \hat{x}_1 > 0 \Rightarrow \hat{\pi}_1 > 0 \Rightarrow \hat{i}_1 > 0.$$

Labour increases because productivity falls more than output:

$$\hat{l}_1 = \hat{y}_1 - \hat{z}_1 = -0.75 - (-1) = 0.25.$$

3. Exercise 8.2: Negative Spending Shock

Tutorial Question

Consider the New Keynesian model with spending shocks described by

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t + u_t^g, \quad (1.NK^{**})$$

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t, \quad (2.NK^{**})$$

$$\hat{i}_t = \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t. \quad (3.NK^{**})$$

with

$$\tilde{\kappa} = (\sigma + \phi) \frac{(1 - \kappa)(1 - \beta\kappa)}{\kappa},$$

$$u_t = E_t[\hat{y}_{t+1}^f] - \hat{y}_t^f, \quad u_t^g = \frac{1}{\sigma} (g_t - E_t[g_{t+1}]),$$

$$v_t = \varepsilon_t, \quad \hat{z}_t = \omega_t,$$

$$\hat{y}_t^f = \left(\frac{1 + \phi}{\sigma + \phi} \right) \hat{z}_t + \left(\frac{1}{\sigma + \phi} \right) g_t.$$

Let the economy be at steady state in period 0. Assume that there is an unanticipated negative spending shock in period 1, i.e.

$$\omega_1^g = -1,$$

and that there are no other shocks in the economy. Let

$$\beta = 0.75, \quad \sigma = \phi = 1, \quad \delta_\pi = 2, \quad \delta_x = 0.5, \quad \rho_g = 0, \quad \kappa = 0.5.$$

All expectations coincide with true values except at period 1. The shock dies off at period 2, so

$$E_1[\hat{x}_2] = \hat{x}_2 = 0, \quad E_1[\hat{\pi}_2] = \hat{\pi}_2 = 0.$$

Compute

$$\hat{i}_1, \quad \hat{x}_1, \quad \hat{\pi}_1, \quad \hat{y}_1, \quad \hat{l}_1.$$

Solution

For these parameter values, the three New Keynesian equations become

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \hat{i}_t + E_t[\hat{\pi}_{t+1}] + u_t + u_t^g,$$

$$\hat{\pi}_t = 0.75E_t[\hat{\pi}_{t+1}] + 1.25\hat{x}_t,$$

$$\hat{i}_t = 2\hat{\pi}_t + 0.5\hat{x}_t.$$

There are no monetary or productivity shocks, so

$$v_t = 0, \quad \hat{z}_t = 0.$$

The spending shock follows

$$g_t = \rho_g g_{t-1} + \omega_t^g.$$

Since the economy starts at steady state,

$$g_0 = 0.$$

Given

$$\rho_g = 0, \quad \omega_1^g = -1,$$

we get

$$g_1 = 0(0) + (-1) = -1.$$

Because $\rho_g = 0$, the shock dies off in period 2, so

$$E_1[g_2] = 0.$$

Flexible-price output is

$$\hat{y}_t^f = \left(\frac{1 + \phi}{\sigma + \phi} \right) \hat{z}_t + \left(\frac{1}{\sigma + \phi} \right) g_t.$$

Since there is no productivity shock,

$$\hat{z}_1 = 0.$$

Since $\sigma = \phi = 1$,

$$\hat{y}_1^f = \frac{1}{2}g_1.$$

Substitute $g_1 = -1$:

$$\hat{y}_1^f = \frac{1}{2}(-1) = -\frac{1}{2}.$$

Since the shock dies off in period 2,

$$E_1[\hat{y}_2^f] = 0.$$

Now compute u_1 :

$$u_1 = E_1[\hat{y}_2^f] - \hat{y}_1^f.$$

Substitute:

$$u_1 = 0 - \left(-\frac{1}{2}\right) = \frac{1}{2}.$$

Now compute the spending shock term:

$$u_1^g = g_1 - E_1[g_2].$$

Substitute:

$$u_1^g = -1 - 0 = -1.$$

In period 2, all variables are at steady state:

$$\hat{x}_2 = \hat{\pi}_2 = \hat{i}_2 = 0.$$

Therefore, in period 1,

$$E_1[\hat{x}_2] = 0, \quad E_1[\hat{\pi}_2] = 0.$$

Use the IS curve:

$$\hat{x}_1 = E_1[\hat{x}_2] - \hat{i}_1 + E_1[\hat{\pi}_2] + u_1 + u_1^g.$$

Substitute:

$$\hat{x}_1 = 0 - \hat{i}_1 + 0 + \frac{1}{2} - 1.$$

Therefore,

$$\hat{x}_1 = -\hat{i}_1 - \frac{1}{2}.$$

Use the Phillips curve:

$$\hat{\pi}_1 = 0.75E_1[\hat{\pi}_2] + 1.25\hat{x}_1.$$

Since $E_1[\hat{\pi}_2] = 0$,

$$\hat{\pi}_1 = 1.25\hat{x}_1.$$

Substitute $\hat{x}_1 = -\hat{i}_1 - \frac{1}{2}$:

$$\hat{\pi}_1 = 1.25 \left(-\hat{i}_1 - \frac{1}{2} \right).$$

Use the Taylor rule:

$$\hat{i}_1 = 2\hat{\pi}_1 + 0.5\hat{x}_1.$$

Substitute $\hat{\pi}_1 = 1.25 \left(-\hat{i}_1 - \frac{1}{2} \right)$ and $\hat{x}_1 = -\hat{i}_1 - \frac{1}{2}$:

$$\begin{aligned}\hat{i}_1 &= 2 \left[1.25 \left(-\hat{i}_1 - \frac{1}{2} \right) \right] + 0.5 \left(-\hat{i}_1 - \frac{1}{2} \right) \\ &= 2.5 \left(-\hat{i}_1 - \frac{1}{2} \right) + 0.5 \left(-\hat{i}_1 - \frac{1}{2} \right) \\ &= 3 \left(-\hat{i}_1 - \frac{1}{2} \right) \\ &= -3\hat{i}_1 - \frac{3}{2}.\end{aligned}$$

Therefore,

$$4\hat{i}_1 = -\frac{3}{2}.$$

So

$$\hat{i}_1 = -\frac{3}{8} = -0.375.$$

Now recover the output gap:

$$\hat{x}_1 = -\hat{i}_1 - \frac{1}{2} = -(-0.375) - 0.5 = 0.375 - 0.5 = -0.125.$$

Inflation is

$$\hat{\pi}_1 = 1.25\hat{x}_1 = 1.25(-0.125) = -0.15625.$$

Actual output is

$$\hat{y}_1 = \hat{x}_1 + \hat{y}_1^f.$$

Substitute:

$$\hat{y}_1 = -0.125 - \frac{1}{2} = -0.625.$$

There is no productivity shock, so

$$\hat{z}_1 = 0.$$

Therefore, labour is

$$\hat{l}_1 = \hat{y}_1 - \hat{z}_1 = -0.625 - 0 = -0.625.$$

Therefore,

$\hat{i}_1 = -0.375,$	$\hat{x}_1 = -0.125,$	$\hat{\pi}_1 = -0.15625,$	$\hat{y}_1 = -0.625,$	$\hat{l}_1 = -0.625.$
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Economic Interpretation

The negative spending shock reduces demand:

$$g_1 = -1.$$

The shock enters the IS curve directly through

$$u_1^g = -1.$$

It also lowers flexible-price output:

$$\hat{y}_1^f = -\frac{1}{2}.$$

Actual output is

$$\hat{y}_1 = -0.625.$$

Since

$$-0.625 < -0.5,$$

actual output falls by more than flexible-price output. Therefore, the output gap is negative:

$$\hat{x}_1 = \hat{y}_1 - \hat{y}_1^f = -0.625 - (-0.5) = -0.125 < 0.$$

A negative output gap lowers inflation:

$$\hat{\pi}_1 = -0.15625 < 0.$$

Since inflation and the output gap are both negative, the Taylor rule implies that the nominal interest rate falls:

$$\hat{i}_1 = -0.375 < 0.$$

$$\text{Spending} \downarrow \Rightarrow \text{Demand} \downarrow \Rightarrow \hat{x}_1 < 0 \Rightarrow \hat{\pi}_1 < 0 \Rightarrow \hat{i}_1 < 0.$$

Because productivity does not change, output and labour move together:

$$\hat{l}_1 = \hat{y}_1 = -0.625.$$

4. Exercise 8.3: Negative Cost-Push Shock

Tutorial Question

Consider the New Keynesian model with a cost-push shock described by

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t, \quad (1.NK^{***})$$

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t + \varepsilon_t^w, \quad (2.NK^{***})$$

$$\hat{i}_t = \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t. \quad (3.NK^{***})$$

with

$$\tilde{\kappa} = (\sigma + \phi) \frac{(1 - \kappa)(1 - \beta\kappa)}{\kappa},$$

$$\varepsilon_t^w = \rho_w \varepsilon_{t-1}^w + \varepsilon_t^o, \quad u_t = E_t[\hat{y}_{t+1}^f] - \hat{y}_t^f,$$

$$v_t = \varepsilon_t, \quad \hat{z}_t = \omega_t, \quad \hat{y}_t^f = \left(\frac{1 + \phi}{\sigma + \phi} \right) \hat{z}_t.$$

Let the economy be at steady state in period 0. Assume that there is an unanticipated negative cost-push shock in period 1, i.e.

$$\varepsilon_1^o = -1,$$

and that there are no other shocks. Let

$$\beta = 0.75, \quad \sigma = \phi = 1, \quad \delta_\pi = 2, \quad \delta_x = 0.5, \quad \rho_w = 0, \quad \kappa = 0.5.$$

All expectations coincide with true values except at period 1. The shock dies off at period 2, so

$$E_1[\hat{x}_2] = \hat{x}_2 = 0, \quad E_1[\hat{\pi}_2] = \hat{\pi}_2 = 0.$$

Compute

$$\hat{i}_1, \quad \hat{x}_1, \quad \hat{\pi}_1, \quad \hat{y}_1, \quad \hat{l}_1.$$

Solution

For these parameter values, the three New Keynesian equations become

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \hat{i}_t + E_t[\hat{\pi}_{t+1}] + u_t,$$

$$\hat{\pi}_t = 0.75E_t[\hat{\pi}_{t+1}] + 1.25\hat{x}_t + \varepsilon_t^w,$$

$$\hat{i}_t = 2\hat{\pi}_t + 0.5\hat{x}_t.$$

There are no monetary shocks, so

$$v_t = 0.$$

There are no productivity shocks, so

$$\hat{z}_t = 0.$$

Therefore, flexible-price output is unchanged:

$$\hat{y}_1^f = 0.$$

This also means

$$u_1 = 0.$$

The cost-push shock follows

$$\varepsilon_t^w = \rho_w \varepsilon_{t-1}^w + \varepsilon_t^o.$$

Since the economy starts at steady state,

$$\varepsilon_0^w = 0.$$

Given

$$\rho_w = 0, \quad \varepsilon_1^o = -1,$$

we get

$$\varepsilon_1^w = 0(0) + (-1) = -1.$$

So

$$\varepsilon_1^w = -1.$$

In period 2, all variables are at steady state:

$$\hat{x}_2 = \hat{\pi}_2 = \hat{i}_2 = 0.$$

Therefore, in period 1,

$$E_1[\hat{x}_2] = 0, \quad E_1[\hat{\pi}_2] = 0.$$

Use the IS curve:

$$\hat{x}_1 = E_1[\hat{x}_2] - \hat{i}_1 + E_1[\hat{\pi}_2] + u_1.$$

Substitute:

$$\hat{x}_1 = 0 - \hat{i}_1 + 0 + 0.$$

Therefore,

$$\hat{x}_1 = -\hat{i}_1.$$

Use the Phillips curve with the cost-push shock:

$$\hat{\pi}_1 = 0.75E_1[\hat{\pi}_2] + 1.25\hat{x}_1 + \varepsilon_1^w.$$

Substitute:

$$\hat{\pi}_1 = 0 + 1.25\hat{x}_1 - 1.$$

Therefore,

$$\hat{\pi}_1 = 1.25\hat{x}_1 - 1.$$

Use the Taylor rule:

$$\hat{i}_1 = 2\hat{\pi}_1 + 0.5\hat{x}_1.$$

Substitute $\hat{\pi}_1 = 1.25\hat{x}_1 - 1$:

$$\begin{aligned} \hat{i}_1 &= 2(1.25\hat{x}_1 - 1) + 0.5\hat{x}_1 \\ &= 2.5\hat{x}_1 - 2 + 0.5\hat{x}_1 \\ &= 3\hat{x}_1 - 2. \end{aligned}$$

Since

$$\hat{x}_1 = -\hat{i}_1,$$

substitute this into the Taylor rule:

$$\hat{i}_1 = 3(-\hat{i}_1) - 2.$$

So

$$\hat{i}_1 = -3\hat{i}_1 - 2.$$

Therefore,

$$4\hat{i}_1 = -2.$$

Hence,

$$\hat{i}_1 = -0.5.$$

Now recover the output gap:

$$\hat{x}_1 = -\hat{i}_1 = -(-0.5) = 0.5.$$

Inflation is

$$\hat{\pi}_1 = 1.25\hat{x}_1 - 1.$$

Substitute:

$$\hat{\pi}_1 = 1.25(0.5) - 1 = 0.625 - 1 = -0.375.$$

Actual output is

$$\hat{y}_1 = \hat{x}_1 + \hat{y}_1^f.$$

Since $\hat{y}_1^f = 0$,

$$\hat{y}_1 = 0.5 + 0 = 0.5.$$

There is no productivity shock, so

$$\hat{z}_1 = 0.$$

Therefore, labour is

$$\hat{l}_1 = \hat{y}_1 - \hat{z}_1 = 0.5 - 0 = 0.5.$$

Therefore,

$$\hat{i}_1 = -0.5, \quad \hat{x}_1 = 0.5, \quad \hat{\pi}_1 = -0.375, \quad \hat{y}_1 = 0.5, \quad \hat{l}_1 = 0.5.$$

Economic Interpretation

The negative cost-push shock directly lowers inflation:

$$\varepsilon_1^w = -1.$$

Even though the output gap becomes positive,

$$\hat{x}_1 = 0.5 > 0,$$

the direct negative cost-push shock dominates. Therefore, inflation is below steady state:

$$\hat{\pi}_1 = -0.375 < 0.$$

Since inflation falls, the Taylor rule implies that the nominal interest rate falls:

$$\hat{i}_1 = -0.5 < 0.$$

The lower interest rate stimulates demand, so the output gap becomes positive:

$$\hat{x}_1 = 0.5 > 0.$$

There is no productivity shock, so flexible-price output is unchanged:

$$\hat{y}_1^f = 0.$$

Therefore,

$$\hat{y}_1 = \hat{x}_1 + \hat{y}_1^f = 0.5.$$

$$\varepsilon_1^w < 0 \Rightarrow \hat{\pi}_1 < 0 \Rightarrow \hat{i}_1 < 0 \Rightarrow \text{Demand } \uparrow \Rightarrow \hat{x}_1 > 0 \Rightarrow \hat{y}_1 > 0.$$

Because productivity does not change, output and labour move together:

$$\hat{l}_1 = \hat{y}_1 = 0.5.$$

5. Memorable Conclusions for Tutorial 8

5.1 Technology Shocks

A technology shock changes productivity and flexible-price output. For a positive technology shock, productivity rises and flexible-price output rises. However, flexible-price output may rise more than actual output because prices are sticky and demand may not fully accommodate the productivity increase.

$$\text{Positive productivity shock} \Rightarrow \hat{z}_t \uparrow \Rightarrow \hat{y}_t^f \uparrow \Rightarrow \hat{x}_t \downarrow \Rightarrow \hat{\pi}_t \downarrow \Rightarrow \hat{i}_t \downarrow.$$

For the negative productivity shock in Tutorial 8:

$$\text{Negative productivity shock} \Rightarrow \hat{z}_1 < 0 \Rightarrow \hat{y}_1^f < 0.$$

Because actual output falls by less than flexible-price output,

$$\hat{y}_1 > \hat{y}_1^f,$$

the output gap becomes positive:

$$\hat{x}_1 > 0.$$

This raises inflation and interest rates:

$$\hat{x}_1 > 0 \Rightarrow \hat{\pi}_1 > 0 \Rightarrow \hat{i}_1 > 0.$$

5.2 Spending Shocks

A spending shock is a demand shock. It changes households' desire to consume in the current period.

A positive spending shock increases demand:

$$\text{Positive spending shock} \Rightarrow \text{Demand} \uparrow \Rightarrow \hat{x}_t \uparrow \Rightarrow \hat{\pi}_t \uparrow \Rightarrow \hat{i}_t \uparrow.$$

For the negative spending shock in Tutorial 8:

$$\text{Negative spending shock} \Rightarrow \text{Demand} \downarrow \Rightarrow \hat{x}_1 < 0 \Rightarrow \hat{\pi}_1 < 0 \Rightarrow \hat{i}_1 < 0.$$

Since there is no productivity shock, output and labour move together:

$$\hat{l}_1 = \hat{y}_1.$$

5.3 Cost-Push Shocks

A cost-push shock affects inflation directly through the Phillips curve:

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t + \varepsilon_t^w.$$

This makes it different from technology and spending shocks.

Cost-push shock $\Rightarrow \hat{\pi}_t$ changes directly $\Rightarrow \hat{i}_t$ responds $\Rightarrow \hat{x}_t$ changes through demand.

For the negative cost-push shock in Tutorial 8:

$$\varepsilon_1^w < 0 \Rightarrow \hat{\pi}_1 < 0 \Rightarrow \hat{i}_1 < 0 \Rightarrow \hat{x}_1 > 0.$$

The output gap becomes positive because the fall in the interest rate stimulates demand.

Memorable Tip

General lesson:

- Technology shocks mainly work through productivity and flexible-price output.
- Spending shocks mainly work through demand.
- Cost-push shocks directly affect inflation.
- Monetary policy responds through the Taylor rule.
- Always identify where the shock enters the model before solving.

6. What-if Questions

6.1 What if the shock were persistent?

In Tutorial 8, the shocks die off in period 2, so

$$E_1[\hat{x}_2] = 0, \quad E_1[\hat{\pi}_2] = 0.$$

If the shock were persistent, expected future output gap and expected future inflation may not be zero. Then period 1 outcomes would change because expectations enter both the IS curve and the Phillips curve.

6.2 What if prices were fully flexible?

If prices were fully flexible, actual output would equal flexible-price output:

$$\hat{y}_t = \hat{y}_t^f.$$

Then the output gap would be zero:

$$\hat{x}_t = 0.$$

This means sticky prices are crucial for explaining why shocks create output gaps and inflation movements in the New Keynesian model.